

MPU-9250

Register Map and Descriptions

Revision 1.4

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MPU-9250 Register Map and Descriptions

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1 Revision History

Revision Date	Revision	Description
9/9/2013	1.4	Initial release

2 Purpose and Scope

This document provides preliminary information regarding the register map and descriptions for the Motion Processing Unit™ MPU-9250™. This document should be used in conjunction with the MPU-9250 Product Specification (PS-MPU-9250A-00) for detailed features, specifications, and other product information.

3 Register Map for Gyroscope and Accelerometer

The following table lists the register map for the gyroscope and accelerometer in the MPU-9250 MotionTracking device.

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
00	0	SELF_TEST_X_GYRO	R/W	xg_st_data [7:0]							
01	1	SELF_TEST_Y_GYRO	R/W	yg_st_data [7:0]							
02	2	SELF_TEST_Z_GYRO	R/W	zg_st_data [7:0]							
0D	13	SELF_TEST_X_ACCEL	R/W	XA_ST_DATA [7:0]							
0E	14	SELF_TEST_Y_ACCEL	R/W	YA_ST_DATA [7:0]							
0F	15	SELF_TEST_Z_ACCEL	R/W	ZA_ST_DATA [7:0]							
13	19	XG_OFFSET_H	R/W	X_OFFSETS_USR [15:8]							
14	20	XG_OFFSET_L	R/W	X_OFFSETS_USR [7:0]							
15	21	YG_OFFSET_H	R/W	Y_OFFSETS_USR [15:8]							
16	22	YG_OFFSET_L	R/W	Y_OFFSETS_USR [7:0]							
17	23	ZG_OFFSET_H	R/W	Z_OFFSETS_USR [15:8]							
18	24	ZG_OFFSET_L	R/W	Z_OFFSETS_USR [7:0]							
19	25	SMPLRT_DIV	R/W	SMPLRT_DIV[7:0]							
1A	26	CONFIG	R/W	-	FIFO MODE	EXT_SYNC_SET[2:0]			DLPF_CFG[2:0]		
1B	27	GYRO_CONFIG	R/W	XGYRO_Ct_en	YGYRO_Ct_en	ZGYRO_Ct_en	GYRO_FS_SEL [1:0]		-	FCHOICE_B[1:0]	
1C	28	ACCEL_CONFIG	R/W	ax_st_en	ay_st_en	az_st_en	ACCEL_FS_SEL[1:0]		-		
1D	29	ACCEL_CONFIG 2	R/W	-				ACCEL_FCHOICE_B		A_DLPF_CFG	
1E	30	LP_ACCEL_ODR	R/W	-				Lposc_clkssel [3:0]			
1F	31	WOM_THR	R/W	WOM_Threshold [7:0]							
23	35	FIFO_EN	R/W	TEMP_FIFO_EN	GYRO_XO_UT	GYRO_YO_UT	GYRO_ZO_UT	ACCEL	SLV2	SLV1	SLV0
24	36	I2C_MST_CTRL	R/W	MULT_MST_EN	WAIT_FOR_ES	SLV_3_FIFO_EN	I2C_MST_P_NSR	I2C_MST_CLK[3:0]			
25	37	I2C_SLV0_ADDR	R/W	I2C_SLV0_RNW	I2C_ID_0 [6:0]						
26	38	I2C_SLV0_REG	R/W	I2C_SLV0_REG[7:0]							
27	39	I2C_SLV0_CTRL	R/W	I2C_SLV0_EN	I2C_SLV0_BYTE_SW	I2C_SLV0_REG_DIS	I2C_SLV0_GRP	I2C_SLV0 LENG[3:0]			
28	40	I2C_SLV1_ADDR	R/W	I2C_SLV1_RNW	I2C_ID_1 [6:0]						
29	41	I2C_SLV1_REG	R/W	I2C_SLV1_REG[7:0]							
2A	42	I2C_SLV1_CTRL	R/W	I2C_SLV1_EN	I2C_SLV1_BYTE_SW	I2C_SLV1_REG_DIS	I2C_SLV1_GRP	I2C_SLV1 LENG[3:0]			
2B	43	I2C_SLV2_ADDR	R/W	I2C_SLV2_RNW	I2C_ID_2 [6:0]						
2C	44	I2C_SLV2_REG	R/W	I2C_SLV2_REG[7:0]							
2D	45	I2C_SLV2_CTRL	R/W	I2C_SLV2_EN	I2C_SLV2_BYTE_SW	I2C_SLV2_REG_DIS	I2C_SLV2_GRP	I2C_SLV2 LENG[3:0]			
2E	46	I2C_SLV3_ADDR	R/W	I2C_SLV3_RNW	I2C_ID_3 [6:0]						
2F	47	I2C_SLV3_REG	R/W	I2C_SLV3_REG[7:0]							
30	48	I2C_SLV3_CTRL	R/W	I2C_SLV3_EN	I2C_SLV3_BYTE_SW	I2C_SLV3_REG_DIS	I2C_SLV3_GRP	I2C_SLV3 LENG [3:0]			
31	49	I2C_SLV4_ADDR	R/W	I2C_SLV4_RNW	I2C_ID_4 [6:0]						
32	50	I2C_SLV4_REG	R/W	I2C_SLV4_REG[7:0]							
33	51	I2C_SLV4_DO	R/W	I2C_SLV4_DO[7:0]							
34	52	I2C_SLV4_CTRL	R/W	I2C_SLV4_EN	SLV4_DON_E_INT_EN	I2C_SLV4_REG_DIS	I2C_MST_DLY[4:0]				

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
35	53	I2C_SLV4_DI	R	I2C_SLV4_DI[7:0]							
36	54	I2C_MST_STATUS	R	PASS_THROUGH	I2C_SLV4_DONE	I2C_LOST_ARB	I2C_SLV4_NACK	I2C_SLV3_NACK	I2C_SLV2_NACK	I2C_SLV1_NACK	I2C_SLV0_NACK
37	55	INT_PIN_CFG	R/W	ACTL	OPEN	LATCH_INT_EN	INT_ANYR_D_2CLEAR	ACTL_FSYNC	FSYNC_INT_MOD_E_EN	BYPASS_EN	-
38	56	INT_ENABLE	R/W	-	WOM_EN	-	FIFO_OFLOW_EN	FSYNC_INT_EN	-	-	RAW_RDY_EN
3A	58	INT_STATUS	R	-	WOM_INT	-	FIFO_OFLOW_INT	FSYNC_INT	-	-	RAW_DATA_RDY_INT
3B	59	ACCEL_XOUT_H	R	ACCEL_XOUT_H[15:8]							
3C	60	ACCEL_XOUT_L	R	ACCEL_XOUT_L[7:0]							
3D	61	ACCEL_YOUT_H	R	ACCEL_YOUT_H[15:8]							
3E	62	ACCEL_YOUT_L	R	ACCEL_YOUT_L[7:0]							
3F	63	ACCEL_ZOUT_H	R	ACCEL_ZOUT_H[15:8]							
40	64	ACCEL_ZOUT_L	R	ACCEL_ZOUT_L[7:0]							
41	65	TEMP_OUT_H	R	TEMP_OUT_H[15:8]							
42	66	TEMP_OUT_L	R	TEMP_OUT_L[7:0]							
43	67	GYRO_XOUT_H	R	GYRO_XOUT_H[15:8]							
44	68	GYRO_XOUT_L	R	GYRO_XOUT_L[7:0]							
45	69	GYRO_YOUT_H	R	GYRO_YOUT_H[15:8]							
46	70	GYRO_YOUT_L	R	GYRO_YOUT_L[7:0]							
47	71	GYRO_ZOUT_H	R	GYRO_ZOUT_H[15:8]							
48	72	GYRO_ZOUT_L	R	GYRO_ZOUT_L[7:0]							
49	73	EXT_SENS_DATA_00	R	EXT_SENS_DATA_00[7:0]							
4A	74	EXT_SENS_DATA_01	R	EXT_SENS_DATA_01[7:0]							
4B	75	EXT_SENS_DATA_02	R	EXT_SENS_DATA_02[7:0]							
4C	76	EXT_SENS_DATA_03	R	EXT_SENS_DATA_03[7:0]							
4D	77	EXT_SENS_DATA_04	R	EXT_SENS_DATA_04[7:0]							
4E	78	EXT_SENS_DATA_05	R	EXT_SENS_DATA_05[7:0]							
4F	79	EXT_SENS_DATA_06	R	EXT_SENS_DATA_06[7:0]							
50	80	EXT_SENS_DATA_07	R	EXT_SENS_DATA_07[7:0]							
51	81	EXT_SENS_DATA_08	R	EXT_SENS_DATA_08[7:0]							
52	82	EXT_SENS_DATA_09	R	EXT_SENS_DATA_09[7:0]							
53	83	EXT_SENS_DATA_10	R	EXT_SENS_DATA_10[7:0]							
54	84	EXT_SENS_DATA_11	R	EXT_SENS_DATA_11[7:0]							
55	85	EXT_SENS_DATA_12	R	EXT_SENS_DATA_12[7:0]							
56	86	EXT_SENS_DATA_13	R	EXT_SENS_DATA_13[7:0]							
57	87	EXT_SENS_DATA_14	R	EXT_SENS_DATA_14[7:0]							
58	88	EXT_SENS_DATA_15	R	EXT_SENS_DATA_15[7:0]							
59	89	EXT_SENS_DATA_16	R	EXT_SENS_DATA_16[7:0]							
5A	90	EXT_SENS_DATA_17	R	EXT_SENS_DATA_17[7:0]							
5B	91	EXT_SENS_DATA_18	R	EXT_SENS_DATA_18[7:0]							
5C	92	EXT_SENS_DATA_19	R	EXT_SENS_DATA_19[7:0]							
5D	93	EXT_SENS_DATA_20	R	EXT_SENS_DATA_20[7:0]							
5E	94	EXT_SENS_DATA_21	R	EXT_SENS_DATA_21[7:0]							
5F	95	EXT_SENS_DATA_22	R	EXT_SENS_DATA_22[7:0]							
60	96	EXT_SENS_DATA_23	R	EXT_SENS_DATA_23[7:0]							
63	99	I2C_SLV0_DO	R/W	I2C_SLV0_DO[7:0]							
64	100	I2C_SLV1_DO	R/W	I2C_SLV1_DO[7:0]							
65	101	I2C_SLV2_DO	R/W	I2C_SLV2_DO[7:0]							

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
66	102	I2C_SLV3_DO	R/W	I2C_SLV3_DO[7:0]									
67	103	I2C_MST_DELAY_CTRL	R/W	DELAY_ES_SHADOW	-	-	I2C_SLV4_DLY_EN	I2C_SLV3_DLY_EN	I2C_SLV2_DLY_EN	I2C_SLV1_DLY_EN	I2C_SLV0_DLY_EN		
68	104	SIGNAL_PATH_RESET	R/W	-	-	-	-	-	GYRO_RST	ACCEL_RST	TEMP_RST		
69	105	MOT_DETECT_CTRL	R/W	ACCEL_INT_EL_EN	ACCEL_INT_EL_MODE	-		-		-			
6A	106	USER_CTRL	R/W	-	FIFO_EN	I2C_MST_EN	I2C_IF_DIS	-	FIFO_RST	I2C_MST_RST	SIG_COND_RST		
6B	107	PWR_MGMT_1	R/W	H_RESET	SLEEP	CYCLE	GYRO_STANDBY	PD_PTAT	CLKSEL[2:0]				
6C	108	PWR_MGMT_2	R/W	-		DIS_XA	DIS_YA	DIS_ZA	DIS_XG	DIS_YG	DIS_ZG		
72	114	FIFO_COUNTH	R/W	-			FIFO_CNT[12:8]						
73	115	FIFO_COUNTL	R/W	FIFO_CNT[7:0]									
74	116	FIFO_R_W	R/W	D[7:0]									
75	117	WHO_AM_I	R	WHOAMI[7:0]									
77	119	XA_OFFSET_H	R/W	XA_OFFS [14:7]									
78	120	XA_OFFSET_L	R/W	XA_OFFS [6:0]									
7A	122	YA_OFFSET_H	R/W	YA_OFFS [14:7]									
7B	123	YA_OFFSET_L	R/W	YA_OFFS [6:0]									
7D	125	ZA_OFFSET_H	R/W	ZA_OFFS [14:7]									
7E	126	ZA_OFFSET_L	R/W	ZA_OFFS [6:0]									

Table 1 MPU-9250 mode register map for Gyroscope and Accelerometer

Note: Register Names ending in _H and _L contain the high and low bytes, respectively, of an internal register value.

In the detailed register tables that follow, register names are in capital letters, while register values are in capital letters and italicized. For example, the ACCEL_XOUT_H register (Register 59) contains the 8 most significant bits, *ACCEL_XOUT*[15:8], of the 16-bit X-Axis accelerometer measurement, *ACCEL_XOUT*.

The reset value is 0x00 for all registers other than the registers below.

- Register 107 (0x01) Power Management 1
- Register 117 (0x71) WHO_AM_I

4 Register Descriptions

This section describes the function and contents of each register within the MPU-9250. All the descriptions relate to the default MPU-9250 mode of operation.

4.1 Registers 0 to 2 – Gyroscope Self-Test Registers

Serial IF: R/W

Reset value: 0x00

REGISTER	BITS	FUNCTION
SELF_TEST_X_GYRO	XG_ST_DATA[7:0]	The value in this register indicates the self test output generated during manufacturing tests. This value is to be used to check against subsequent self test outputs performed by the end user.
SELF_TEST_Y_GYRO	YG_ST_DATA[7:0]	The value in this register indicates the self test output generated during manufacturing tests. This value is to be used to check against subsequent self test outputs performed by the end user.
SELF_TEST_Z_GYRO	ZG_ST_DATA[7:0]	The value in this register indicates the self test output generated during manufacturing tests. This value is to be used to check against subsequent self test outputs performed by the end user.

For details of the MPU-9250 self-test implementation, please refer to the following document: AN-MPU-9250A-03, MPU-9250 Accelerometer, Gyroscope and Compass Self-Test Implementation.

4.2 Registers 13 to 15 – Accelerometer Self-Test Registers

Serial IF: R/W

Reset value: 0x00

REGISTER	BITS	FUNCTION
SELF_TEST_X_ACCEL	XA_ST_DATA[7:0]	The value in this register indicates the self test output generated during manufacturing tests. This value is to be used to check against subsequent self test outputs performed by the end user.
SELF_TEST_Y_ACCEL	YA_ST_DATA[7:0]	The value in this register indicates the self test output generated during manufacturing tests. This value is to be used to check against subsequent self test outputs performed by the end user.
SELF_TEST_Z_ACCEL	ZA_ST_DATA[7:0]	The value in this register indicates the self test output generated during manufacturing tests. This value is to be used to check against subsequent self test outputs performed by the end user.

For details of the MPU-9250 self-test implementation, please refer to the following document: AN-MPU-9250A-03, MPU-9250 Accelerometer, Gyroscope and Compass Self-Test Implementation.

4.3 Registers 19 to 24 – Gyro Offset Registers

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	X_OFFS_USR[15:8]	High byte, Low byte in USR register (14h) OffsetLSB= $X_OFFS_USR * 4 / 2^{FS_SEL}$ OffsetDPS= $X_OFFS_USR * 4 / 2^{FS_SEL} / Gyro_Sensitivity$ Nominal $FS_SEL = 0$ Conditions $Gyro_Sensitivity = 2^{16} \text{ LSB} / 500\text{dps}$ Max 999.969 dps Min -1000 dps Step 0.0305 dps
[7:0]	X_OFFS_USR[7:0]	Low byte, High byte in USR register (13h)
[7:0]	Y_OFFS_USR[15:8]	High byte, Low byte in USR register (16h) OffsetLSB= $Y_OFFS_USR * 4 / 2^{FS_SEL}$ OffsetDPS= $Y_OFFS_USR * 4 / 2^{FS_SEL} / Gyro_Sensitivity$ Nominal $FS_SEL = 0$ Conditions $Gyro_Sensitivity = 2^{16} \text{ LSB} / 500\text{dps}$ Max 999.969 dps Min -1000 dps Step 0.0305 dps
[7:0]	Y_OFFS_USR[7:0]	Low byte, High byte in USR register (15h)
[7:0]	Z_OFFS_USR[15:8]	High byte, Low byte in USR register (18h) OffsetLSB= $Z_OFFS_USR * 4 / 2^{FS_SEL}$ OffsetDPS= $Z_OFFS_USR * 4 / 2^{FS_SEL} / Gyro_Sensitivity$ Nominal $FS_SEL = 0$ Conditions $Gyro_Sensitivity = 2^{16} \text{ LSB} / 500\text{dps}$ Max 999.969 dps Min -1000 dps Step 0.0305 dps
[7:0]	Z_OFFS_USR[7:0]	Low byte, High byte in USR register (17h)

These registers are used to remove DC bias from the gyro sensor data output for X, Y and Z axes. The values in these registers are subtracted from the gyro sensor values before going into the sensor registers. Please refer to registers 67 to 72 for units.

4.4 Register 25 – Sample Rate Divider

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	SMPLRT_DIV[7:0]	Divides the internal sample rate (see register CONFIG) to generate the sample rate that controls sensor data output rate, FIFO sample rate. NOTE: This register is only effective when Fchoice = 2'b11 (fchoice_b register bits are 2'b00), and (0 < dlpf_cfg < 7), such that the average filter's output is selected (see chart below). This is the update rate of sensor register. SAMPLE_RATE= Internal_Sample_Rate / (1 + SMPLRT_DIV)

Data should be sampled at or above sample rate; SMPLRT_DIV is only used for 1kHz internal sampling.

4.5 Register 26 – Configuration

BIT	NAME	FUNCTION
[7]	-	Reserved
[6]	FIFO_MODE	When set to '1', when the fifo is full, additional writes will not be written to fifo. When set to '0', when the fifo is full, additional writes will be written to the fifo, replacing the oldest data.

[5:3]	EXT_SYNC_SET[2:0]	Enables the FSYNC pin data to be sampled. <table><tr><th>EXT_SYNC_SET</th><th>FSYNC bit location</th></tr><tr><td>0</td><td>function disabled</td></tr><tr><td>1</td><td>TEMP_OUT_L[0]</td></tr><tr><td>2</td><td>GYRO_XOUT_L[0]</td></tr><tr><td>3</td><td>GYRO_YOUT_L[0]</td></tr><tr><td>4</td><td>GYRO_ZOUT_L[0]</td></tr><tr><td>5</td><td>ACCEL_XOUT_L[0]</td></tr><tr><td>6</td><td>ACCEL_YOUT_L[0]</td></tr><tr><td>7</td><td>ACCEL_ZOUT_L[0]</td></tr></table> Fsync will be latched to capture short strobes. This will be done such that if Fsync toggles, the latched value toggles, but won't toggle again until the new latched value is captured by the sample rate strobe. This is a requirement for working with some 3rd party devices that have fsync strobes shorter than our sample rate.	EXT_SYNC_SET	FSYNC bit location	0	function disabled	1	TEMP_OUT_L[0]	2	GYRO_XOUT_L[0]	3	GYRO_YOUT_L[0]	4	GYRO_ZOUT_L[0]	5	ACCEL_XOUT_L[0]	6	ACCEL_YOUT_L[0]	7	ACCEL_ZOUT_L[0]
EXT_SYNC_SET	FSYNC bit location																			
0	function disabled																			
1	TEMP_OUT_L[0]																			
2	GYRO_XOUT_L[0]																			
3	GYRO_YOUT_L[0]																			
4	GYRO_ZOUT_L[0]																			
5	ACCEL_XOUT_L[0]																			
6	ACCEL_YOUT_L[0]																			
7	ACCEL_ZOUT_L[0]																			
[2:0]	DLPF_CFG[2:0]	For the DLPF to be used, fchoice[1:0] must be set to 2'b11, fchoice_b[1:0] is 2'b00. See table 3 below.																		

The DLPF is configured by **DLPF_CFG**, when **FCHOICE_B** [1:0] = 2b'00. The gyroscope and temperature sensor are filtered according to the value of **DLPF_CFG** and **FCHOICE_B** as shown in the table below. Note that **FCHOICE** mentioned in the table below is the inverted value of **FCHOICE_B** (e.g. **FCHOICE**=2b'00 is same as **FCHOICE_B**=2b'11).

FCHOICE		DLPF_CFG	Gyroscope			Temperature Sensor	
<1>	<0>		Bandwidth (Hz)	Delay (ms)	Fs (kHz)	Bandwidth (Hz)	Delay (ms)
x	0	x	8800	0.064	32	4000	0.04
0	1	x	3600	0.11	32	4000	0.04
1	1	0	250	0.97	8	4000	0.04
1	1	1	184	2.9	1	188	1.9
1	1	2	92	3.9	1	98	2.8
1	1	3	41	5.9	1	42	4.8
1	1	4	20	9.9	1	20	8.3
1	1	5	10	17.85	1	10	13.4
1	1	6	5	33.48	1	5	18.6
1	1	7	3600	0.17	8	4000	0.04

4.6 Register 27 – Gyroscope Configuration

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	XGYRO_Cten	X Gyro self-test
[6]	YGYRO_Cten	Y Gyro self-test
[5]	ZGYRO_Cten	Z Gyro self-test
[4:3]	GYRO_FS_SEL[1:0]	Gyro Full Scale Select: 00 = +250dps 01 = +500 dps 10 = +1000 dps 11 = +2000 dps
[2]	-	Reserved
[1:0]	Fchoice_b[1:0]	Used to bypass DLPF as shown in table 1 above. NOTE: Register is Fchoice_b (inverted version of Fchoice), table 1 uses Fchoice (which is the inverted version of this register).

4.7 Register 28 – Accelerometer Configuration

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	ax_st_en	X Accel self-test
[6]	ay_st_en	Y Accel self-test
[5]	az_st_en	Z Accel self-test
[4:3]	ACCEL_FS_SEL[1:0]	Accel Full Scale Select: ±2g (00), ±4g (01), ±8g (10), ±16g (11)
[2:0]	-	Reserved

4.8 Register 29 – Accelerometer Configuration 2

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:6]	Reserved	
[5:4]	Reserved	
[3]	accel_fchoice_b	Used to bypass DLPF as shown in table 2 below. NOTE: This register contains accel_fchoice_b (the inverted version of accel_fchoice as described in the table below).
[2:0]	A_DLPFCFG	Accelerometer low pass filter setting as shown in table 2 below.

Accelerometer Data Rates and Bandwidths (Normal Mode)

ACCEL_FCHOICE	A_DLPF_CFG	Output			
		Bandwidth (Hz)	Delay (ms)	Noise Density (ug/rtHz)	Rate (kHz)
0	X	1.13 K	0.75	250	4
1	0	460	1.94	250	1
1	1	184	5.80	250	1
1	2	92	7.80	250	1
1	3	41	11.80	250	1
1	4	20	19.80	250	1
1	5	10	35.70	250	1
1	6	5	66.96	250	1
1	7	460	1.94	250	1

The data output rate of the DLPF filter block can be further reduced by a factor of $1/(1+SMPLRT_DIV)$, where SMPLRT_DIV is an 8-bit integer. Following is a small subset of ODRs that are configurable for the accelerometer in the normal mode in this manner (Hz):

3.91, 7.81, 15.63, 31.25, 62.50, 125, 250, 500, 1K

The following table lists the approximate accelerometer filter bandwidths available in the low-power mode of operation.

In the low-power mode of operation, the accelerometer is duty-cycled. Fchoice=0 for all options.

Accelerometer Data Rates and Bandwidths (Low-Power Mode)

ACCEL_FCHOICE	ODR (Hz)	Output	
		Bandwidth (Hz)	Delay (ms)
0	0.24	1.1 k	1
0	0.49	1.1 k	1
0	0.98	1.1 k	1
0	1.95	1.1 k	1

0	3.91	1.1 k	1
0	7.81	1.1 k	1
0	15.63	1.1 k	1
0	31.25	1.1 k	1
0	62.50	1.1 k	1
0	125	1.1 k	1
0	250	1.1 k	1
0	500	1.1 kHz	1

As you can see from the tables above, some of the ODRs can be configured in the normal accelerometer mode as well as low power mode.

For further details on how to configure the individual ODRs, please refer to register 30 Low Power Accelerometer ODR Control.

4.9 Register 30 – Low Power Accelerometer ODR Control

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION																												
[7:4]	Reserved																													
[3:0]	lposc_clkssel[3:0]	Sets the frequency of waking up the chip to take a sample of accel data – the low power accel Output Data Rate. <table><tr><th>Lposc_clkssel</th><th>Output Frequency (Hz)</th></tr><tr><td>0</td><td>0.24</td></tr><tr><td>1</td><td>0.49</td></tr><tr><td>2</td><td>0.98</td></tr><tr><td>3</td><td>1.95</td></tr><tr><td>4</td><td>3.91</td></tr><tr><td>5</td><td>7.81</td></tr><tr><td>6</td><td>15.63</td></tr><tr><td>7</td><td>31.25</td></tr><tr><td>8</td><td>62.50</td></tr><tr><td>9</td><td>125</td></tr><tr><td>10</td><td>250</td></tr><tr><td>11</td><td>500</td></tr><tr><td>12-15</td><td>RESERVED</td></tr></table>	Lposc_clkssel	Output Frequency (Hz)	0	0.24	1	0.49	2	0.98	3	1.95	4	3.91	5	7.81	6	15.63	7	31.25	8	62.50	9	125	10	250	11	500	12-15	RESERVED
Lposc_clkssel	Output Frequency (Hz)																													
0	0.24																													
1	0.49																													
2	0.98																													
3	1.95																													
4	3.91																													
5	7.81																													
6	15.63																													
7	31.25																													
8	62.50																													
9	125																													
10	250																													
11	500																													
12-15	RESERVED																													

4.10 Register 31 – Wake-on Motion Threshold

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	WOM_Threshold	This register holds the threshold value for the Wake on Motion Interrupt for accel x/y/z axes. LSB = 4mg. Range is 0mg to 1020mg.

For more details on how to configure the Wake-on-Motion interrupt, please refer to section 5 in the MPU-9250 Product Specification document.

4.11 Register 35 – FIFO Enable

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	TEMP_OUT	1 – Write TEMP_OUT_H and TEMP_OUT_L to the FIFO at the sample rate; If enabled, buffering of data occurs even if data path is in standby. 0 – function is disabled
[6]	GYRO_XOUT	1 – Write GYRO_XOUT_H and GYRO_XOUT_L to the FIFO at the sample rate; If enabled, buffering of data occurs even if data path is in standby. 0 – function is disabled
[5]	GYRO_YOUT	1 – Write GYRO_YOUT_H and GYRO_YOUT_L to the FIFO at the sample rate; If enabled, buffering of data occurs even if data path is in standby. 0 – function is disabled NOTE: Enabling any one of the bits corresponding to the Gyros or Temp data paths, data is buffered into the FIFO even though that data path is not enabled.
[4]	GYRO_ZOUT	1 – Write GYRO_ZOUT_H and GYRO_ZOUT_L to the FIFO at the sample rate; If enabled, buffering of data occurs even if data path is in standby. 0 – function is disabled
[3]	ACCEL	1 – write ACCEL_XOUT_H, ACCEL_XOUT_L, ACCEL_YOUT_H, ACCEL_YOUT_L, ACCEL_ZOUT_H, and ACCEL_ZOUT_L to the FIFO at the sample rate; 0 – function is disabled

BIT	NAME	FUNCTION
[2]	SLV_2	1 – write EXT_SENS_DATA registers associated to SLV_2 (as determined by I2C_SLV0_CTRL, I2C_SLV1_CTRL, and I2C_SLV2_CTRL) to the FIFO at the sample rate; 0 – function is disabled
[1]	SLV_1	1 – write EXT_SENS_DATA registers associated to SLV_1 (as determined by I2C_SLV0_CTRL and I2C_SLV1_CTRL) to the FIFO at the sample rate; 0 – function is disabled
[0]	SLV_0	1 – write EXT_SENS_DATA registers associated to SLV_0 (as determined by I2C_SLV0_CTRL) to the FIFO at the sample rate; 0 – function is disabled NOTE: See I2C_SLV3_CTRL register to enable this feature for SLV_3

Note: For further information regarding the association of EXT_SENS_DATA registers to particular slave devices, please refer to Registers 73 to 96.

4.12 Register 36 – I2C Master Control

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	MULT_MST_EN	Enables multi-master capability. When disabled, clocking to the I2C_MST_IF can be disabled when not in use and the logic to detect lost arbitration is disabled.
[6]	WAIT_FOR_ES	Delays the data ready interrupt until external sensor data is loaded. If I2C_MST_IF is disabled, the interrupt will still occur.
[5]	SLV_3_FIFO_EN	1 – write EXT_SENS_DATA registers associated to SLV_3 (as determined by I2C_SLV0_CTRL and I2C_SLV1_CTRL and I2C_SLV2_CTRL) to the FIFO at the sample rate; 0 – function is disabled
[4]	I2C_MST_P_NSR	This bit controls the I2C Master's transition from one slave read to the next slave read. If 0, there is a restart between reads. If 1, there is a stop between reads.

BIT	NAME	FUNCTION																																																			
[3:0]	I2C_MST_CLK [3:0]	<i>I2C_MST_CLK</i> is a 4 bit unsigned value which configures a divider on the MPU-9250 internal 8MHz clock. It sets the I²C master clock speed according to the following table: <table data-bbox="730 420 1271 972"> <thead> <tr> <th>I2C_MST_CLK</th><th>I²C Master Clock Speed</th><th>8MHz Clock Divider</th></tr> </thead> <tbody> <tr><td>0</td><td>348 kHz</td><td>23</td></tr> <tr><td>1</td><td>333 kHz</td><td>24</td></tr> <tr><td>2</td><td>320 kHz</td><td>25</td></tr> <tr><td>3</td><td>308 kHz</td><td>26</td></tr> <tr><td>4</td><td>296 kHz</td><td>27</td></tr> <tr><td>5</td><td>286 kHz</td><td>28</td></tr> <tr><td>6</td><td>276 kHz</td><td>29</td></tr> <tr><td>7</td><td>267 kHz</td><td>30</td></tr> <tr><td>8</td><td>258 kHz</td><td>31</td></tr> <tr><td>9</td><td>500 kHz</td><td>16</td></tr> <tr><td>10</td><td>471 kHz</td><td>17</td></tr> <tr><td>11</td><td>444 kHz</td><td>18</td></tr> <tr><td>12</td><td>421 kHz</td><td>19</td></tr> <tr><td>13</td><td>400 kHz</td><td>20</td></tr> <tr><td>14</td><td>381 kHz</td><td>21</td></tr> <tr><td>15</td><td>364 kHz</td><td>22</td></tr> </tbody> </table>	I2C_MST_CLK	I ² C Master Clock Speed	8MHz Clock Divider	0	348 kHz	23	1	333 kHz	24	2	320 kHz	25	3	308 kHz	26	4	296 kHz	27	5	286 kHz	28	6	276 kHz	29	7	267 kHz	30	8	258 kHz	31	9	500 kHz	16	10	471 kHz	17	11	444 kHz	18	12	421 kHz	19	13	400 kHz	20	14	381 kHz	21	15	364 kHz	22
I2C_MST_CLK	I ² C Master Clock Speed	8MHz Clock Divider																																																			
0	348 kHz	23																																																			
1	333 kHz	24																																																			
2	320 kHz	25																																																			
3	308 kHz	26																																																			
4	296 kHz	27																																																			
5	286 kHz	28																																																			
6	276 kHz	29																																																			
7	267 kHz	30																																																			
8	258 kHz	31																																																			
9	500 kHz	16																																																			
10	471 kHz	17																																																			
11	444 kHz	18																																																			
12	421 kHz	19																																																			
13	400 kHz	20																																																			
14	381 kHz	21																																																			
15	364 kHz	22																																																			

Note: For further information regarding the association of EXT_SENS_DATA registers to particular slave devices, please refer to Registers 73 to 96.

4.13 Registers 37 to 39 – I²C Slave 0 Control

Register 37 - I2C_SLV0_ADDR

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV0_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_0[6:0]	Physical address of I2C slave 0

Register 38 - I2C_SLV0_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV0_REG[7:0]	I2C slave 0 register address from where to begin data transfer

Register 39 - I2C_SLV0_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV0_EN	1 – Enable reading data from this slave at the sample rate and storing data at the first available EXT_SENS_DATA register, which is always EXT_SENS_DATA_00 for I2C slave 0. 0 – function is disabled for this slave

BIT	NAME	FUNCTION
[6]	I2C_SLV0_BYTE_SW	1 – Swap bytes when reading both the low and high byte of a word. Note there is nothing to swap after reading the first byte if I2C_SLV0_REG[0] = 1, or if the last byte read has a register address lsb = 0. For example, if I2C_SLV0_REG = 0x1, and I2C_SLV0_LEN = 0x4: 1) The first byte read from address 0x1 will be stored at EXT_SENS_DATA_00, 2) the second and third bytes will be read and swapped, so the data read from address 0x2 will be stored at EXT_SENS_DATA_02, and the data read from address 0x3 will be stored at EXT_SENS_DATA_01, 3) The last byte read from address 0x4 will be stored at EXT_SENS_DATA_03 0 – no swapping occurs, bytes are written in order read.
[5]	I2C_SLV0_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data
[4]	I2C_SLV0_GRP	External sensor data typically comes in as groups of two bytes. This bit is used to determine if the groups are from the slave's register address 0 and 1, 2 and 3, etc., or if the groups are address 1 and 2, 3 and 4, etc.. 0 indicates slave register addresses 0 and 1 are grouped together (odd numbered register ends the group). 1 indicates slave register addresses 1 and 2 are grouped together (even numbered register ends the group). This allows byte swapping of registers that are grouped starting at any address.
[3:0]	I2C_SLV0_LEN[3:0]	Number of bytes to be read from I2C slave 0

4.14 Registers 40 to 42 – I²C Slave 1 Control

Register 40 - I2C_SLV1_ADDR

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV1_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_1[6:0]	Physical address of I2C slave 1

Register 41 - I2C_SLV1_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV1_REG[7:0]	I2C slave 1 register address from where to begin data transfer

Register 42 - I2C_SLV1_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV1_EN	1 – Enable reading data from this slave at the sample rate and storing data at the first available EXT_SENS_DATA register as determined by I2C_SLV1_EN and I2C_SLV1 LENG. 0 – function is disabled for this slave

BIT	NAME	FUNCTION
[6]	I2C_SLV1_BYTE_SW	1 – Swap bytes when reading both the low and high byte of a word. Note there is nothing to swap after reading the first byte if I2C_SLV1_REG[0] = 1, or if the last byte read has a register address lsb = 0. For example, if I2C_SLV1_EN = 0x1, and I2C_SLV1 LENG = 0x3 (to show swap has to do with I2C slave address not EXT_SENS_DATA address), and if I2C_SLV1_REG = 0x1, and I2C_SLV1 LENG = 0x4: 1) The first byte read from address 0x1 will be stored at EXT_SENS_DATA_03 (slave 0's data will be in EXT_SENS_DATA_00, EXT_SENS_DATA_01, and EXT_SENS_DATA_02), 2) the second and third bytes will be read and swapped, so the data read from address 0x2 will be stored at EXT_SENS_DATA_04, and the data read from address 0x3 will be stored at EXT_SENS_DATA_05, 3) The last byte read from address 0x4 will be stored at EXT_SENS_DATA_06 0 – no swapping occurs, bytes are written in order read.
[5]	I2C_SLV1_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data
[4]	I2C_SLV1_GRP	External sensor data typically comes in as groups of two bytes. This bit is used to determine if the groups are from the slave's register address 0 and 1, 2 and 3, etc., or if the groups are address 1 and 2, 3 and 4, etc.. 0 indicates slave register addresses 0 and 1 are grouped together (odd numbered register ends the group). 1 indicates slave register addresses 1 and 2 are grouped together (even numbered register ends the group). This allows byte swapping of registers that are grouped starting at any address.
[3:0]	I2C_SLV1_LENG[3:0]	Number of bytes to be read from I2C slave 1

4.15 Registers 43 to 45 – I²C Slave 2 Control

Register 43 - I2C_SLV2_ADDR

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV2_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_2[6:0]	Physical address of I2C slave 2

Register 44 - I2C_SLV2_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV2_REG[7:0]	I2C slave 2 register address from where to begin data transfer

Register 45 - I2C_SLV2_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV2_EN	1 – Enable reading data from this slave at the sample rate and storing data at the first available EXT_SENS_DATA register as determined by I2C_SLV0_EN, I2C_SLV0 LENG, I2C_SLV1_EN and I2C_SLV1 LENG. 0 – function is disabled for this slave
[6]	I2C_SLV2_BYTE_SW	1 – Swap bytes when reading both the low and high byte of a word. Note there is nothing to swap after reading the first byte if I2C_SLV2_REG[0] = 1, or if the last byte read has a register address lsb = 0. See I2C_SLV1_CTRL for an example. 0 – no swapping occurs, bytes are written in order read.
[5]	I2C_SLV2_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data

BIT	NAME	FUNCTION
[4]	I2C_SLV2_GRP	External sensor data typically comes in as groups of two bytes. This bit is used to determine if the groups are from the slave's register address 0 and 1, 2 and 3, etc., or if the groups are address 1 and 2, 3 and 4, etc.. 0 indicates slave register addresses 0 and 1 are grouped together (odd numbered register ends the group). 1 indicates slave register addresses 1 and 2 are grouped together (even numbered register ends the group). This allows byte swapping of registers that are grouped starting at any address.
[3:0]	I2C_SLV2 LENG[3:0]	Number of bytes to be read from I2C slave 2

4.16 Registers 46 to 48 – I²C Slave 3 Control

Register 46 - I2C_SLV3_ADDR

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV3_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_3[6:0]	Physical address of I2C slave 3

Register 47 - I2C_SLV3_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV3_REG[7:0]	I2C slave 3 register address from where to begin data transfer

Register 48 - I2C_SLV3_CTRL

Serial IF: R/W

Reset value: 0x00

BITS	NAME	FUNCTION
[7]	I2C_SLV3_EN	1 – Enable reading data from this slave at the sample rate and storing data at the first available EXT_SENS_DATA register as determined by I2C_SLV0_EN, I2C_SLV0_LEN, I2C_SLV1_EN, I2C_SLV1_LEN, I2C_SLV2_EN and I2C_SLV2_LEN. 0 – function is disabled for this slave
[6]	I2C_SLV3_BYTE_SW	1 – Swap bytes when reading both the low and high byte of a word. Note there is nothing to swap after reading the first byte if I2C_SLV3_REG[0] = 1, or if the last byte read has a register address lsb = 0. See I2C_SLV1_CTRL for an example. 0 – no swapping occurs, bytes are written in order read.
[5]	I2C_SLV0_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data
[4]	I2C_SLV3_GRP	External sensor data typically comes in as groups of two bytes. This bit is used to determine if the groups are from the slave's register address 0 and 1, 2 and 3, etc., or if the groups are address 1 and 2, 3 and 4, etc.. 0 indicates slave register addresses 0 and 1 are grouped together (odd numbered register ends the group). 1 indicates slave register addresses 1 and 2 are grouped together (even numbered register ends the group). This allows byte swapping of registers that are grouped starting at any address.
[3:0]	I2C_SLV3_LEN[3:0]	Number of bytes to be read from I2C slave 3

4.17 Registers 49 to 53 – I²C Slave 4 Control

Register 49 - I2C_SLV4_ADDR

Serial IF: R/W

Reset value: 0x00

BITS	NAME	FUNCTION
[7]	I2C_SLV4_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_4[6:0]	Physical address of I2C slave 4

Register 50 - I2C_SLV4_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV4_REG[7:0]	I2C slave 4 register address from where to begin data transfer

Register 51 - I2C_SLV4_DO**Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:0]	I2C_SLV4_DO[7:0]	Data to be written to I2C Slave 4 if enabled.

Register 52 - I2C_SLV4_CTRL**Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7]	I2C_SLV4_EN	1 – Enable data transfer with this slave at the sample rate. If read command, store data in I2C_SLV4_DI register, if write command, write data stored in I2C_SLV4_DO register. Bit is cleared when a single transfer is complete. Be sure to write I2C_SLV4_DO first 0 – function is disabled for this slave
[6]	SLV4_DONE_INT_EN	1 – Enables the completion of the I2C slave 4 data transfer to cause an interrupt. 0 – Completion of the I2C slave 4 data transfer will not cause an interrupt.
[5]	I2C_SLV4_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data
[4:0]	I2C_MST_DLY	When enabled via the I2C_MST_DELAY_CTRL, those slaves will only be enabled every (1+I2C_MST_DLY) samples (as determined by the SMPLRT_DIV and DLPF_CFG registers.

Register 53 - I2C_SLV4_DI**Serial IF: R****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:0]	I2C_SLV4_DI[7:0]	Data read from I2C Slave 4.

4.18 Register 54 – I²C Master Status

Serial IF: R/C

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	PASS_THROUGH	Status of FSYNC interrupt – used as a way to pass an external interrupt through this chip to the host. If enabled in the INT_PIN_CFG register by asserting bit FSYNC_INT_EN and if the FSYNC signal transitions from low to high, this will cause an interrupt. A read of this register clears all status bits in this register.
[6]	I2C_SLV4_DONE	Asserted when I2C slave 4's transfer is complete, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted, and if the SLV4_DONE_INT_EN bit is asserted in the I2C_SLV4_CTRL register.
[5]	I2C_LOST_ARB	Asserted when I2C slave loses arbitration of the I2C bus, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[4]	I2C_SLV4_NACK	Asserted when slave 4 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[3]	I2C_SLV3_NACK	Asserted when slave 3 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[2]	I2C_SLV2_NACK	Asserted when slave 2 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[1]	I2C_SLV1_NACK	Asserted when slave 1 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[0]	I2C_SLV0_NACK	Asserted when slave 0 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.

4.19 Register 55 – INT Pin / Bypass Enable Configuration

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	ACTL	1 – The logic level for INT pin is active low. 0 – The logic level for INT pin is active high.
[6]	OPEN	1 – INT pin is configured as open drain. 0 – INT pin is configured as push-pull.
[5]	LATCH_INT_EN	1 – INT pin level held until interrupt status is cleared. 0 – INT pin indicates interrupt pulse's is width 50us.
[4]	INT_ANYRD_2CLEAR	1 – Interrupt status is cleared if any read operation is performed. 0 – Interrupt status is cleared only by reading INT_STATUS register
[3]	ACTL_FSYNC	1 – The logic level for the FSYNC pin as an interrupt is active low. 0 – The logic level for the FSYNC pin as an interrupt is active high.
[2]	FSYNC_INT_MODE_EN	1 – This enables the FSYNC pin to be used as an interrupt. A transition to the active level described by the ACTL_FSYNC bit will cause an interrupt. The status of the interrupt is read in the I2C Master Status register PASS_THROUGH bit. 0 – This disables the FSYNC pin from causing an interrupt.
[1]	BYPASS_EN	When asserted, the i2c_master interface pins(ES_CL and ES_DA) will go into 'bypass mode' when the i2c master interface is disabled. The pins will float high due to the internal pull-up if not enabled and the i2c master interface is disabled.
[0]	RESERVED	

4.20 Register 56 – Interrupt Enable

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	RESERVED	

BIT	NAME	FUNCTION
[6]	WOM_EN	1 – Enable interrupt for wake on motion to propagate to interrupt pin. 0 – function is disabled.
[5]	RESERVED	
[4]	FIFO_OVERFLOW_EN	1 – Enable interrupt for fifo overflow to propagate to interrupt pin. 0 – function is disabled.
[3]	FSYNC_INT_EN	1 – Enable Fsync interrupt to propagate to interrupt pin. 0 – function is disabled.
[2]	RESERVED	
[1]	RESERVED	
[0]	RAW_RDY_EN	1 – Enable Raw Sensor Data Ready interrupt to propagate to interrupt pin. The timing of the interrupt can vary depending on the setting in register 36 I2C_MST_CTRL, bit [6] WAIT_FOR_ES. 0 – function is disabled.

4.21 Register 58 – Interrupt Status

Serial IF: R/C

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	Reserved	
[6]	WOM_INT	1 – Wake on motion interrupt occurred.
[5]	Reserved	
[4]	FIFO_OVERFLOW_INT	1 – Fifo Overflow interrupt occurred. Note that the oldest data is has been dropped from the fifo.
[3]	FSYNC_INT	1 – Fsync interrupt occurred.
[2]	Reserved	
[1]	Reserved	
[0]	RAW_DATA_RDY_INT	1 – Sensor Register Raw Data sensors are updated and Ready to be read. The timing of the interrupt can vary depending on the setting in register 36 I2C_MST_CTRL, bit [6] WAIT_FOR_ES.

4.22 Registers 59 to 64 – Accelerometer Measurements**Name:** ACCEL_XOUT_H

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	D[7:0]	High byte of accelerometer x-axis data.

Name: ACCEL_XOUT_L

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	D[7:0]	Low byte of accelerometer x-axis data.

Name: ACCEL_YOUT_H

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	D[7:0]	High byte of accelerometer y-axis data.

Name: ACCEL_YOUT_L

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	D[7:0]	Low byte of accelerometer y-axis data.

Name: ACCEL_ZOUT_H

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	D[7:0]	High byte of accelerometer z-axis data.

Name: ACCEL_ZOUT_L

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	D[7:0]	Low byte of accelerometer z-axis data.

4.23 Registers 65 and 66 – Temperature Measurement

Name: TEMP_OUT_H

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
[7:0]	D[7:0]	High byte of the temperature sensor output

Name: TEMP_OUT_L

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
[7:0]	D[7:0]	Low byte of the temperature sensor output: $\text{TEMP_degC} = ((\text{TEMP_OUT} - \text{RoomTemp_Offset}) / \text{Temp_Sensitivity}) + 21\text{degC}$ Where Temp_degC is the temperature in degrees C measured by the temperature sensor. TEMP_OUT is the actual output of the temperature sensor.

4.24 Registers 67 to 72 – Gyroscope Measurements

Name: GYRO_XOUT_H

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
[7:0]	D[7:0]	High byte of the X-Axis gyroscope output

Name: GYRO_XOUT_L

Serial IF: SyncR

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
------	------	----------

BIT	NAME	FUNCTION
[7:0]	D[7:0]	Low byte of the X-Axis gyroscope output GYRO_XOUT = Gyro_Sensitivity * X_angular_rate Nominal FS_SEL = 0 Conditions Gyro_Sensitivity = 131 LSB/(°/s)

Name: GYRO_YOUT_H**Serial IF: SyncR****Reset value: 0x00 (if sensor disabled)**

BIT	NAME	FUNCTION
[7:0]	D[7:0]	High byte of the Y-Axis gyroscope output

Name: GYRO_YOUT_L**Serial IF: SyncR****Reset value: 0x00 (if sensor disabled)**

BIT	NAME	FUNCTION
[7:0]	D[7:0]	Low byte of the Y-Axis gyroscope output GYRO_YOUT = Gyro_Sensitivity * Y_angular_rate Nominal FS_SEL = 0 Conditions Gyro_Sensitivity = 131 LSB/(°/s)

Name: GYRO_ZOUT_H**Serial IF: SyncR****Reset value: 0x00 (if sensor disabled)**

BIT	NAME	FUNCTION
[7:0]	D[7:0]	High byte of the Z-Axis gyroscope output

Name: GYRO_ZOUT_L**Serial IF: SyncR****Reset value: 0x00 (if sensor disabled)**

BIT	NAME	FUNCTION
[7:0]	D[7:0]	Low byte of the Z-Axis gyroscope output GYRO_ZOUT = Gyro_Sensitivity * Z_angular_rate Nominal FS_SEL = 0 Conditions Gyro_Sensitivity = 131 LSB/(°/s)

4.25 Registers 73 to 96 – External Sensor Data

EXT_SENS_DATA_00 – 23

Serial IF: SyncR

Reset value: 0x00

24 registers with the same description as below:

BIT	NAME	FUNCTION
[7:0]	D[7:0]	Sensor data read from external I2C devices via the I2C master interface. The data stored is controlled by the I2C_SLV(0-4)_ADDR, I2C_SLV(0-4)_REG, and I2C_SLV(0-4)_CTRL registers

Description:

These registers store data read from external sensors by the Slave 0, 1, 2, and 3 on the auxiliary I²C interface. Data read by Slave 4 is stored in I2C_SLV4_DI (Register 53).

External sensor data is written to these registers at the Sample Rate as defined in Register 25. This access rate can be reduced by using the Slave Delay Enable registers (Register 103).

Data is placed in these external sensor data registers according to I2C_SLV0_CTRL, I2C_SLV1_CTRL, I2C_SLV2_CTRL, and I2C_SLV3_CTRL (Registers 39, 42, 45, and 48). When more than zero bytes are read ($I2C_SLVx_LEN > 0$) from an enabled slave ($I2C_SLVx_EN = 1$), the slave is read at the Sample Rate (as defined in Register 25) or delayed rate (if specified in Register 52 and 103). During each sample cycle, slave reads are performed in order of Slave number. If all slaves are enabled with more than zero bytes to be read, the order will be Slave 0, followed by Slave 1, Slave 2, and Slave 3.

Each enabled slave will have EXT_SENS_DATA registers associated with it by number of bytes read ($I2C_SLVx_LEN$) in order of slave number, starting from EXT_SENS_DATA_00. Note that this means enabling or disabling a slave may change the higher numbered slaves' associated registers. Furthermore, if fewer total bytes are being read from the external sensors as a result of such a change, then the data remaining in the registers which no longer have an associated slave device (i.e. high numbered registers) will remain in these previously allocated registers unless reset.

If the sum of the read lengths of all SLVx transactions exceed the number of available EXT_SENS_DATA registers, the excess bytes will be dropped. There are 24 EXT_SENS_DATA

registers and hence the total read lengths between all the slaves cannot be greater than 24 or some bytes will be lost.

Note: Slave 4's behavior is distinct from that of Slaves 0-3. For further information regarding the characteristics of Slave 4, please refer to Registers 49 to 53.

Example:

Suppose that Slave 0 is enabled with 4 bytes to be read ($I2C_SLV0_EN = 1$ and $I2C_SLV0_LEN = 4$) while Slave 1 is enabled with 2 bytes to be read, ($I2C_SLV1_EN=1$ and $I2C_SLV1_LEN = 2$). In such a situation, `EXT_SENS_DATA_00` through `_03` will be associated with Slave 0, while `EXT_SENS_DATA_04` and `05` will be associated with Slave 1.

If Slave 2 is enabled as well, registers starting from `EXT_SENS_DATA_06` will be allocated to Slave 2.

If Slave 2 is disabled while Slave 3 is enabled in this same situation, then registers starting from `EXT_SENS_DATA_06` will be allocated to Slave 3 instead.

Register Allocation for Dynamic Disable vs. Normal Disable

If a slave is disabled at any time, the space initially allocated to the slave in the `EXT_SENS_DATA` register, will remain associated with that slave. This is to avoid dynamic adjustment of the register allocation.

The allocation of the `EXT_SENS_DATA` registers is recomputed only when (1) all slaves are disabled, or (2) the `I2C_MST_RST` bit is set (Register 106).

This above is also true if one of the slaves gets NACKed and stops functioning.

4.26 Register 99 – I²C Slave 0 Data Out**I2C_SLV0_DO****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:0]	I2C_SLV0_DO	Data out when slave 0 is set to write

For further information regarding Slave 1 control, please refer to Registers 37 to 39.

4.27 Register 100 – I²C Slave 1 Data Out**I2C_SLV1_DO****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:0]	I2C_SLV1_DO	Data out when slave 1 is set to write

For further information regarding Slave 1 control, please refer to Registers 40 to 42.

4.28 Register 101 – I²C Slave 2 Data Out**I2C_SLV2_DO****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:0]	I2C_SLV2_DO	Data out when slave 2 is set to write

For further information regarding Slave 2 control, please refer to Registers 43 to 45.

4.29 Register 102 – I²C Slave 3 Data Out

I2C_SLV3_DO

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV3_DO	Data out when slave 3 is set to write

For further information regarding Slave 2 control, please refer to Registers 46 to 48.

4.30 Register 103 – I²C Master Delay Control
I2C_MST_DELAY_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	DELAY_ES_SHADOW	Delays shadowing of external sensor data until all data is received
[6:5]	Reserved	
[4]	I2C_SLV4_DLY_EN	When enabled, slave 4 will only be accessed (1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG
[3]	I2C_SLV3_DLY_EN	When enabled, slave 3 will only be accessed (1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG
[2]	I2C_SLV2_DLY_EN	When enabled, slave 2 will only be accessed 1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG
[1]	I2C_SLV1_DLY_EN	When enabled, slave 1 will only be accessed 1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG
[0]	I2C_SLV0_DLY_EN	When enabled, slave 0 will only be accessed 1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG

4.31 Register 104 – Signal Path Reset**SIGNAL_PATH_RESET****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:3]	Reserved	
[2]	GYRO_RST	Reset gyro digital signal path. Note: Sensor registers are not cleared. Use SIG_COND_RST to clear sensor registers.
[1]	ACCEL_RST	Reset accel digital signal path. Note: Sensor registers are not cleared. Use SIG_COND_RST to clear sensor registers.
[0]	TEMP_RST	Reset temp digital signal path. Note: Sensor registers are not cleared. Use SIG_COND_RST to clear sensor registers.

4.32 Register 105 – Accelerometer Interrupt Control**ACCEL_INTEL_CTRL****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7]	ACCEL_INTEL_EN	This bit enables the Wake-on-Motion detection logic.
[6]	ACCEL_INTEL_MODE	This bit defines 1 = Compare the current sample with the previous sample. 0 = Not used.
[5:0]	Reserved	

Please refer to the Wake-on-Motion Interrupt section of the MPU-9250 Product Specification for additional details.

4.33 Register 106 – User Control**Name: USER_CTRL****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7]	Reserved	
[6]	FIFO_EN	1 – Enable FIFO operation mode. 0 – Disable FIFO access from serial interface. To disable FIFO writes by dma, use FIFO_EN register. To disable possible FIFO writes from DMP, disable the DMP.
[5]	I2C_MST_EN	1 – Enable the I2C Master I/F module; pins ES_DA and ES_SCL are isolated from pins SDA/SDI and SCL/ SCLK. 0 – Disable I2C Master I/F module; pins ES_DA and ES_SCL are logically driven by pins SDA/SDI and SCL/ SCLK. NOTE: DMP will run when enabled, even if all internal sensors are disabled, except when the sample rate is set to 8Khz.
[4]	I2C_IF_DIS	1 – Reset I2C Slave module and put the serial interface in SPI mode only. This bit auto clears after one clock cycle.
[3]	Reserved	
[2]	FIFO_RST	1 – Reset FIFO module. Reset is asynchronous. This bit auto clears after one clock cycle.
[1]	I2C_MST_RST	1 – Reset I2C Master module. Reset is asynchronous. This bit auto clears after one clock cycle. NOTE: This bit should only be set when the I2C master has hung. If this bit is set during an active I2C master transaction, the I2C slave will hang, which will require the host to reset the slave.
[0]	SIG_COND_RST	1 – Reset all gyro digital signal path, accel digital signal path, and temp digital signal path. This bit also clears all the sensor registers. SIG_COND_RST is a pulse of one clk8M wide.

4.34 Register 107 – Power Management 1

Name: PWR_MGMT_1

Serial IF: R/W

Reset value: (Depends on PU_SLEEP_MODE bit, see below)

BIT	NAME	FUNCTION
[7]	H_RESET	1 – Reset the internal registers and restores the default settings. Write a 1 to set the reset, the bit will auto clear.
[6]	SLEEP	When set, the chip is set to sleep mode (After OTP loads, the PU_SLEEP_MODE bit will be written here)

BIT	NAME	FUNCTION
[5]	CYCLE	When set, and SLEEP and STANDBY are not set, the chip will cycle between sleep and taking a single sample at a rate determined by LP_ACCEL_ODR register NOTE: When all accelerometer axis are disabled via PWR_MGMT_2 register bits and cycle is enabled, the chip will wake up at the rate determined by the respective registers above, but will not take any samples.
[4]	GYRO_STANDBY	When set, the gyro drive and pll circuitry are enabled, but the sense paths are disabled. This is a low power mode that allows quick enabling of the gyros.
[3]	PD_PTAT	Power down internal PTAT voltage generator and PTAT ADC
[2:0]	CLKSEL[2:0]	Code Clock Source 0 Internal 20MHz oscillator 1 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 2 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 3 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 4 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 5 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 6 Internal 20MHz oscillator 7 Stops the clock and keeps timing generator in reset (After OTP loads, the inverse of PU_SLEEP_MODE bit will be written to CLKSEL[0])

4.35 Register 108 – Power Management 2

Name: PWR_MGMT_2

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:6]	Reserved	

BIT	NAME	FUNCTION
[5]	DISABLE_XA	1 – X accelerometer is disabled 0 – X accelerometer is on
[4]	DISABLE_YA	1 – Y accelerometer is disabled 0 – Y accelerometer is on
[3]	DISABLE_ZA	1 – Z accelerometer is disabled 0 – Z accelerometer is on
[2]	DISABLE_XG	1 – X gyro is disabled 0 – X gyro is on
[1]	DISABLE_YG	1 – Y gyro is disabled 0 – Y gyro is on
[0]	DISABLE_ZG	1 – Z gyro is disabled 0 – Z gyro is on

The MPU-9250 can be put into Accelerometer Only Low Power Mode using the following steps:

- (i) Set CYCLE bit to 1
- (ii) Set SLEEP bit to 0
- (iii) Set TEMP_DIS bit to 1
- (iv) Set DIS_XG, DIS_YG, DIS_ZG bits to 1

The bits mentioned in the steps (i) to (iii) can be found in Power Management 1 register (Register 107).

In this mode, the device will power off all devices except for the primary I²C interface, waking only the accelerometer at fixed intervals to take a single measurement.

4.36 Register 114 and 115 – FIFO Count Registers

Name: FIFO_COUNTH

Address: 114

Serial IF: Read Only

Reset value: 0x00

BIT	NAME	FUNCTION
[7:5]	Reserved	

BIT	NAME	FUNCTION
[4:0]	FIFO_CNT[12:8]	High Bits, count indicates the number of written bytes in the FIFO. Reading this byte latches the data for both FIFO_COUNTH, and FIFO_COUNTL.

FIFO_COUNTL
Address: 115
Serial IF: Read Only
Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	FIFO_CNT[7:0]	Low Bits, count indicates the number of written bytes in the FIFO. NOTE: Must read FIFO_COUNTH to latch new data for both FIFO_COUNTH and FIFO_COUNTL.

4.37 Register 116 – FIFO Read Write
Name: FIFO_R_W
Serial IF: R/W
Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	D[7:0]	Read/Write command provides Read or Write operation for the FIFO.

Description:

This register is used to read and write data from the FIFO buffer.

Data is written to the FIFO in order of register number (from lowest to highest). If all the FIFO enable flags (see below) are enabled and all External Sensor Data registers (Registers 73 to 96) are associated with a Slave device, the contents of registers 59 through 96 will be written in order at the Sample Rate.

The contents of the sensor data registers (Registers 59 to 96) are written into the FIFO buffer when their corresponding FIFO enable flags are set to 1 in FIFO_EN (Register 35). An additional flag for the sensor data registers associated with I²C Slave 3 can be found in I2C_MST_CTRL (Register 36).

If the FIFO buffer has overflowed, the status bit *FIFO_OFLOW_INT* is automatically set to 1. This bit is located in INT_STATUS (Register 58). When the FIFO buffer has overflowed, the oldest data will be lost and new data will be written to the FIFO unless register 26 CONFIG, bit[6] FIFO_MODE = 1.

If the FIFO buffer is empty, reading this register will return the last byte that was previously read from the FIFO until new data is available. The user should check *FIFO_COUNT* to ensure that the FIFO buffer is not read when empty.

4.38 Register 117 – Who Am I

Name: WHOAMI

Serial IF: Read Only

Reset value: 0x68

BIT	NAME	FUNCTION
[7:0]	WHOAMI	Register to indicate to user which device is being accessed.

This register is used to verify the identity of the device. The contents of *WHO_AM_I* is an 8-bit device ID. The default value of the register is 0x71.

4.39 Registers 119, 120, 122, 123, 125, 126 Accelerometer Offset Registers

For MPU-9250 mode:

Name: XA_OFFS_H

Address: 119

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	XA_OFFS[14:7]	Upper bits of the X accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps

Name: XA_OFFS_L

Address: 120

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:1]	XA_OFFS[6:0]	Lower bits of the X accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps

BIT	NAME	FUNCTION
[0]	Reserved	

Name: YA_OFFS_H**Address: 122****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:0]	YA_OFFS[14:7]	Upper bits of the Y accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps

Name: YA_OFFS_L**Address: 123****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:1]	YA_OFFS[6:0]	Lower bits of the Y accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps
[0]	Reserved	

Name: ZA_OFFS_H**Address: 125****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:0]	ZA_OFFS[14:7]	Upper bits of the Z accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps

Name: ZA_OFFS_L**Address: 126****Serial IF: R/W****Reset value: 0x00**

BIT	NAME	FUNCTION
[7:1]	ZA_OFFS[6:0]	Lower bits of the Z accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps
[0]	Reserved	

5 Register Map for Magnetometer

The register map for the MPU-9250's Magnetometer (AK8963) section is listed below.

Name	Address	READ/ WRITE	Description	Bit width	Explanation
WIA	00H	READ	Device ID	8	
INFO	01H	READ	Information	8	
ST1	02H	READ	Status 1	8	Data status
HXL	03H	READ	Measurement data	8	X-axis data
HXH	04H			8	
HYL	05H			8	Y-axis data
HYH	06H			8	
HZL	07H			8	Z-axis data
HZH	08H			8	
ST2	09H	READ	Status 2	8	Data status
CNTL	0AH	READ/ WRITE	Control	8	
RSV	0BH	READ/ WRITE	Reserved	8	DO NOT ACCESS
ASTC	0CH	READ/ WRITE	Self-test	8	
TS1	0DH	READ/ WRITE	Test 1	8	DO NOT ACCESS
TS2	0EH	READ/ WRITE	Test 2	8	DO NOT ACCESS
I2CDIS	0FH	READ/ WRITE	I ² C disable	8	
ASAX	10H	READ	X-axis sensitivity adjustment value	8	Fuse ROM
ASAY	11H	READ	Y-axis sensitivity adjustment value	8	Fuse ROM
ASAZ	12H	READ	Z-axis sensitivity adjustment value	8	Fuse ROM

Table 2 Register Table

Addresses from 00H to 0CH and from 10H to 12H are compliant with automatic increment function of serial interface respectively. Values of addresses from 10H to 12H can be read only in Fuse access mode. In other modes, read data is not correct.

5.1 Register Map Description

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only Register									
00H	WIA	0	1	0	0	1	0	0	0
01H	INFO	INFO7	INFO6	INFO5	INFO4	INFO3	INFO2	INFO1	INFO0
02H	ST1	0	0	0	0	0	0	DOR	DRDY
03H	HXL	HX7	HX6	HX5	HX4	HX3	HX2	HX1	HX0
04H	HXH	HX15	HX14	HX13	HX12	HX11	HX10	HX9	HX8
05H	HYL	HY7	HY6	HY5	HY4	HY3	HY2	HY1	HY0
06H	HYH	HY15	HY14	HY13	HY12	HY11	HY10	HY9	HY8
07H	HZL	HZ7	HZ6	HZ5	HZ4	HZ3	HZ2	HZ1	HZ0
08H	HZH	HZ15	HZ14	HZ13	HZ12	HZ11	HZ10	HZ9	HZ8
09H	ST2	0	0	0	BITM	HOFL	0	0	0
Write/read Register									
0AH	CNTL1	0	0	0	0	MODE3	MODE2	MODE1	MODE0
0BH	CNTL2	0	0	0	0	0	0	0	SRST
0CH	ASTC	-	SELF	-	-	-	-	-	-
0DH	TS1	-	-	-	-	-	-	-	-
0EH	TS2	-	-	-	-	-	-	-	-
0FH	I2CDIS	I2CDIS7	I2CDIS6	I2CDIS5	I2CDIS4	I2CDIS3	I2CDIS2	I2CDIS1	I2CDIS0
Read-only Register									
10H	ASAX	COEFX7	COEFX6	COEFX5	COEFX4	COEFX3	COEFX2	COEFX1	COEFX0
11H	ASAY	COEFY7	COEFY6	COEFY5	COEFY4	COEFY3	COEFY2	COEFY1	COEFY0
12H	ASAZ	COEFZ7	COEFZ6	COEFZ5	COEFZ4	COEFZ3	COEFZ2	COEFZ1	COEFZ0

Table 3 Register Map

Note: When VDD is turned ON, POR function works and all registers of AK893 are initialized.

TS1 and TS2 are test registers for shipment test. Do not use these registers.

RSV is reserved register. Do not use this register.

5.2 Detailed Descriptions for Magnetometer Registers

This section details each register within the MPU-9250's Magnetometer section.

5.3 WIA: Device ID

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
00H	WIA	0	1	0	0	1	0	0	0

Device ID of AKM. It is described in one byte and fixed value.

48H: fixed

5.4 INFO: Information

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
01H	INFO	INFO7	INFO6	INFO5	INFO4	INFO3	INFO2	INFO1	INFO0

INFO[7:0]: Device information for AKM.

5.5 ST1: Status 1

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
02H	ST1	0	0	0	0	0	0	0	DRDY
	Reset	0	0	0	0	0	0	0	0

DRDY: Data Ready

"0": Normal

"1": Data is ready

DRDY bit turns to "1" when data is ready in single measurement mode or self-test mode. It returns to "0" when any one of ST2 register or measurement data register (HXL to HZH) is read.

DOR: Data Overrun

"0": Normal

"1": Data overrun

DOR bit turns to "1" when data has been skipped in continuous measurement mode or external trigger measurement mode. It returns to "0" when any one of ST2 register or measurement data register (HXL~HZH) is read.

5.6 HXL to HZH: Measurement Data

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
03H	HXL	HX7	HX6	HX5	HX4	HX3	HX2	HX1	HX0
04H	HXH	HX15	HX14	HX13	HX12	HX11	HX10	HX9	HX8
05H	HYL	HY7	HY6	HY5	HY4	HY3	HY2	HY1	HY0
06H	HYH	HY15	HY14	HY13	HY12	HY11	HY10	HY9	HY8
07H	HZL	HZ7	HZ6	HZ5	HZ4	HZ3	HZ2	HZ1	HZ0
08H	HZH	HZ15	HZ14	HZ13	HZ12	HZ11	HZ10	HZ9	HZ8
Reset		0	0	0	0	0	0	0	0

Measurement data of magnetic sensor X-axis/Y-axis/Z-axis

HXL[7:0]: X-axis measurement data lower 8bit

HXH[15:8]: X-axis measurement data higher 8bit

HYL[7:0]: Y-axis measurement data lower 8bit

HYH[15:8]: Y-axis measurement data higher 8bit

HZL[7:0]: Z-axis measurement data lower 8bit

HZH[15:8]: Z-axis measurement data higher 8bit

Measurement data is stored in two's complement and Little Endian format. Measurement range of each axis is from -32760 ~ 32760 decimal in 16-bit output.

Measurement data (each axis) [15:0]			Magnetic flux density [μ T]
Two's complement	Hex	Decimal	
0111 1111 1111 1000	7FF8	32760	4912(max.)
0000 0000 0000 0001	0001	1	0.15
0000 0000 0000 0000	0000	0	0
1111 1111 1111 1111	FFFF	-1	-0.15
1000 0000 0000 1000	8008	-32760	-4912(min.)

Table 4 Measurement data format

5.7 ST2: Status 2

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
09H	ST2	0	0	0	BITM	HOFL	0	0	0
	Reset	0	0	0	0	0	0	0	0

HOFL: Magnetic sensor overflow

"0": Normal

"1": Magnetic sensor overflow occurred

In single measurement mode, continuous measurement mode, external trigger measurement mode and self-test mode, magnetic sensor may overflow even though measurement data register is not saturated. In this case, measurement data is not correct and HOFL bit turns to "1". When next measurement starts, it returns to "0".

BITM: Output bit setting (mirror)

"0": 14-bit output

"1": 16-bit output

Mirror data of BIT bit of CNTL1 register.

ST2 register has a role as data reading end register, also. When any of measurement data register is read in continuous measurement mode or external trigger measurement mode, it means data reading start and taken as data reading until ST2 register is read. Therefore, when any of measurement data is read, be sure to read ST2 register at the end.

5.8 CNTL1: Control 1

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
0AH	CNTL1	0	0	0	BIT	MODE3	MODE2	MODE1	MODE0
	Reset	0	0	0	0	0	0	0	0

MODE[3:0]: Operation mode setting

"0000": Power-down mode

"0001": Single measurement mode

"0010": Continuous measurement mode 1

"0110": Continuous measurement mode 2

"0100": External trigger measurement mode

"1000": Self-test mode

"1111": Fuse ROM access mode

Other code settings are prohibited

BIT: Output bit setting

"0": 14-bit output

"1": 16-bit output

When each mode is set, AK8963 transits to set mode.

When CNTL register is accessed to be written, registers from 02H to 09H are initialized.

5.9 CNTL2: Control 2

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
0BH	CNTL2	0	0	0	0	0	0	0	SRST
Reset		0	0	0	0	0	0	0	0

SRST: Soft reset

"0": Normal

"1": Reset

When "1" is set, all registers are initialized. After reset, SRST bit turns to "0" automatically.

5.10 ASTC: Self-Test Control

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Write/read register									
0CH	ASTC	-	SELF	-	-	-	-	-	-
Reset		0	0	0	0	0	0	0	0

SELF: Self-test control

"0": Normal

"1": Generate magnetic field for self-test

Do not write "1" to any bit other than SELF bit in ASTC register. If "1" is written to any bit other than SELF bit, normal measurement cannot be done.

5.11 TS1, TS2: Test 1, 2

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Write/read register									
0DH	TS1	-	-	-	-	-	-	-	-
0EH	TS2	-	-	-	-	-	-	-	-
Reset		0	0	0	0	0	0	0	0

TS1 and TS2 registers are test registers for shipment test. Do not use these registers.

5.12 I2CDIS: I²C Disable

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Write/read register									
0FH	I2CDIS	I2CDIS7	I2CDIS6	I2CDIS5	I2CDIS4	I2CDIS3	I2CDIS2	I2CDIS1	I2CDIS0
Reset		0	0	0	0	0	0	0	0

This register disables I²C bus interface. I²C bus interface is enabled in default. To disable I²C bus interface, write "00011011" to I2CDIS register. Then I²C bus interface is disabled.

Once I²C bus interface is disabled, it is impossible to write other value to I2CDIS register. To enable I2C bus interface, reset AK8963 or input start condition 8 times continuously.

5.13 ASAX, ASAY, ASAZ: Sensitivity Adjustment values

Addr	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
10H	ASAX	COEFX7	COEFX6	COEFX5	COEFX4	COEFX3	COEFX2	COEFX1	COEFX0
11H	ASAY	COEFY7	COEFY6	COEFY5	COEFY4	COEFY3	COEFY2	COEFY1	COEFY0
12H	ASAZ	COEFZ7	COEFZ6	COEFZ5	COEFZ4	COEFZ3	COEFZ2	COEFZ1	COEFZ0
Reset		-	-	-	-	-	-	-	-

Sensitivity adjustment data for each axis is stored to fuse ROM on shipment.

ASAX[7:0]: Magnetic sensor X-axis sensitivity adjustment value

ASAY[7:0]: Magnetic sensor Y-axis sensitivity adjustment value

ASAZ[7:0]: Magnetic sensor Z-axis sensitivity adjustment value

■ Sensitivity Adjustment

The sensitivity adjustment is done by the equation below;

$$H_{adj} = H \times \left(\frac{(ASA - 128) \times 0.5}{128} + 1 \right),$$

where H is the measurement data read out from the measurement data register, ASA is the sensitivity adjustment value, and H_{adj} is the adjusted measurement data.

6 Advanced Hardware Features

The MPU-9250 includes advanced hardware features that support Android that can be enabled and disabled through simple hardware register settings. The advanced hardware features are not initially enabled after device power up, and must be individually enabled and configured. The following motion-based functions are supported and do not require an external hub or microprocessor:

- Android Orientation
- Step Count, Step Detection
- Significant Motion Detection
- Batch mode
- Low Power Quaternion (3, 6, 9 axis)

Features supported for embedded applications include:

- Pedometer, Directional Tap
- Low Power Quaternion (3, 6, 9 axis)

Features supported for Windows 8 UMDF implementation (no external hub required) include:

- Quaternion Output with CS/CSI filtering

For further details please see the Application Note “Programming Sequence for DMP Hardware Functions.”

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